EXCEL MODERN FUNCTIONS

Practical Handbook | Lessons 106-113 | Excel for Microsoft 365

Eight sequential lessons covering TOCOL, TOROW, VSTACK, HSTACK, TAKE, DROP, CHOOSECOLS, CHOOSEROWS and advanced GROUPBY reporting. Includes copy-ready formulas, worked examples, outputs, and common mistakes.

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Remember: These formula-based functions depend on the Excel version. GROUPBY is available in supported Microsoft 365 builds; older perpetual versions may not include it. Sample ranges refer to the teaching datasets, not a supplied workbook.

Lesson 106 | TOCOL

TOCOL flattens a one- or two-dimensional range into one vertical column. By default it scans by row. Its ignore argument determines whether blanks and errors are included.

```
=TOCOL(array,[ignore],[scan_by_column])
```

Argument	Values / behavior
ignore	0 keep everything; 1 ignore blanks; 2 ignore errors; 3 ignore both
scan_by_column	FALSE (default): row-first; TRUE: column-first

Example dataset (B2:F3)

Month	B	C	D	E	F
May	1166	1582	blank	1140	900
June	1035	1232	919	blank	873

Useful formulas and exact results

```
=TOCOL(B2:F2)
```

Returns 1166, 1582, 0, 1140, 900 down a column: blank cells appear as zeros by default.

```
=TOCOL(B2:F2,3)
```

Returns 1166, 1582, 1140, 900. Blank and error positions are omitted.

```
=TOCOL(B2:F3,3)
```

Row-first result: 1166, 1582, 1140, 900, 1035, 1232, 919, 873.

```
=TOCOL(B2:F3,3,TRUE)
```

Column-first result: 1166, 1035, 1582, 1232, 919, 1140, 900, 873.

Remember: TOCOL spills its output from one formula cell. The intended spill range must be empty.

Lesson 107 | TOROW and UNIQUE

TOROW flattens an array into one horizontal row. Its ignore and scan arguments operate like TOCOL.

```
=TOROW(array,[ignore],[scan_by_column])
```

Example customer names across two months

Month	Customer 1	Customer 2	Customer 3	Customer 4	Customer 5
May	Ravi	Priya	blank	Arjun	Sneha
June	Ravi	Neha	Kiran	blank	Priya

```
=TOROW(B2:F3,3)
```

Row-first output: Ravi, Priya, Arjun, Sneha, Ravi, Neha, Kiran, Priya.

```
=TOROW(B2:F3,3,TRUE)
```

Column-first output: Ravi, Ravi, Priya, Neha, Kiran, Arjun, Sneha, Priya.

Remove duplicate names from the horizontal result

```
=UNIQUE(TOROW(B2:F3,3,TRUE),TRUE)
```

Returns six distinct names: Ravi, Priya, Neha, Kiran, Arjun, Sneha. The second argument of UNIQUE must be TRUE because the individual values are columns in a horizontal array.

```
=UNIQUE(TOROW(B2:F3,3,TRUE),TRUE,TRUE)
```

Returns only names appearing exactly once: Neha, Kiran, Arjun, Sneha. The final TRUE means exactly_once.

Remember: UNIQUE(TOROW(...)) without by_col=TRUE treats the whole horizontal row as a single record and does not remove repeated customer names.

Lesson 108 | VSTACK and HSTACK

VSTACK appends datasets as new rows; HSTACK adds datasets as new columns. Both return spilling arrays.

Function	Syntax	Use case
VSTACK	=VSTACK(array1,[array2],...)	Combine joiners from 2023, 2024 and 2025
HSTACK	=HSTACK(array1,[array2],...)	Place basic, contact and employment fields side by side

VSTACK: stack yearly employee records

```
=VSTACK('2023_Joiners'!A1:D4,'2024_Joiners'!A2:D4,'2025_Joiners'!A2:D4)
```

Expected result: one header row plus nine employee records. Select headers only from the first dataset to avoid repeated headers in the combined output.

HSTACK: bring aligned employee columns together

```
=HSTACK(Basic_Info!A1:C4,Contact_Info!B1:C4,Employment_Info!B1:C4)
```

Expected columns: Employee ID, Name, Department, Email, Phone, Job Title, Manager. Repeated Employee ID columns are omitted from the later arrays.

Remember: HSTACK aligns by row position, NOT by Employee ID. If source sheets are sorted differently, match records with XLOOKUP or a proper merge before stacking. Differently sized arrays are padded with #N/A.

Lesson 109 | TAKE, DROP, CHOOSECOLS and CHOOSEROWS

Function	Syntax	Action
TAKE	=TAKE(array,rows,[columns])	Retain top/bottom rows and left/right columns
DROP	=DROP(array,rows,[columns])	Remove top/bottom rows and left/right columns
CHOOSECOLS	=CHOOSECOLS(array,col_num1,...)	Pick and reorder specific columns
CHOOSEROWS	=CHOOSEROWS(array,row_num1,...)	Pick and reorder specific rows

Assume A1:J11 contains a header and ten employee records; columns 1, 2, 6, 7 and 8 are ID, Name, Joining Date, Mobile and Email.

```
=TAKE(A1:J11,5,8)
```

Returns first five rows (one header plus four employees) and first eight columns.

```
=TAKE(A1:J11,-3)
```

Returns the last three employees.

```
=DROP(A1:J11,5)
```

Excludes the header and first four employees; six employee rows remain.

```
=DROP(A1:J11,0,-2)
```

Keeps all rows, dropping the last two columns.

```
=CHOOSECOLS(A1:J11,1,2,7,8,6)
```

Returns ID, Name, Mobile, Email and Joining Date, in that exact order.

```
=CHOOSEROWS(A1:J11,1,3,5,8)
```

Returns array rows 1, 3, 5 and 8 (header plus selected employee records).

Preserving a header when dropping the first rows

```
=VSTACK(TAKE(A1:J11,1),DROP(A1:J11,5))
```

Reattaches the header to the remaining records.

Remember: Negative TAKE/DROP counts operate from the end. CHOOSECOLS and CHOOSEROWS indices refer to positions inside the selected array. These functions select data but do not enforce data-access security.

Lesson 110 | GROUPBY introduction and aggregations

GROUPBY summarizes numerical data for categories using a worksheet formula, similar to a PivotTable. The first three arguments are mandatory.

```
=GROUPBY(row_fields,values,function,[field_headers],[total_depth],[sort_order],[filter_array],[field_relationship])
```

Argument	Meaning
row_fields	One or more grouping category columns
values	One or more columns being summarized
function	Aggregation (SUM, AVERAGE, COUNT, MAX, etc.)

Practice data: product sales and customer ratings

Assume B2:B13 contains Product Category; C2:C13 contains Sales Channel; D2:D13 is Net Sales; and E2:E13 is Customer Rating.

```
=GROUPBY(B2:B13,D2:D13,SUM)
```

Product Category	Total Net Sales
Accessories	8,000
Footwear	16,500
Kids Wear	31,500
Grand Total	56,000

```
=GROUPBY(C2:C13,E2:E13,AVERAGE)
```

Sales Channel	Average Customer Rating
In Store	3.67
Online	4.50
Grand Total	4.08

Remember: Pass SUM or AVERAGE as a function argument, not SUM() or AVERAGE(). GROUPBY recomputes when its referenced source changes; fixed ranges do not automatically include newly added rows.

Lesson 111 | GROUPBY multiple fields, values and headers

GROUPBY supports multiple row grouping fields and multiple value columns. In the practice dataset B=Brand, C=Sales Channel, D=Units Sold, E=Unit Price and F=Net Sales (rows 2:13).

Sum two adjacent value fields

```
=GROUPBY(B2:B13,D2:E13,SUM)
```

Brand	Total Units Sold	Sum of Unit Prices
AlphaWear	14	4,000
BetaStyle	14	2,000
GammaFit	10	6,000
Grand Total	38	12,000

Summing Unit Price is a formula demonstration; it is not generally a meaningful business KPI. Units Sold plus Net Sales would be more useful.

Group by a combination of Brand and Sales Channel

```
=GROUPBY(B2:C13,F2:F13,SUM,2,0)
```

Brand	Channel	Net Sales
AlphaWear	In Store	9,000
AlphaWear	Online	5,000

Brand	Channel	Net Sales
BetaStyle	In Store	2,500
BetaStyle	Online	4,500
GammaFit	In Store	10,500
GammaFit	Online	4,500

Fourth argument: field_headers

Value	Behavior
0	Automatically detect headers
1	Source has headers, do not display
2	Source has no headers, generate headings
3	Source has headers, display them

```
=GROUPBY(B1:B13,D1:E13,SUM,3)
```

The first row is explicitly treated as headers and included in the displayed summary headings.

Group nonadjacent value columns

```
=GROUPBY(B2:B13,HSTACK(D2:D13,F2:F13),SUM)
```

Produces Total Units Sold and Total Net Sales by brand. HSTACK constructs a temporary two-column value array.

Lesson 112 | GROUPBY totals, sorting and filtering

Fifth argument: total_depth

Value	Totals behavior
0	No totals
1	Grand total below groups
2	Grand total and subtotals below groups
-1	Grand total above groups
-2	Grand total and subtotals above groups

```
=GROUPBY(B2:B13,F2:F13,SUM,2,1)
```

Brand totals with overall grand total of 36,000.

```
=GROUPBY(B2:C13,F2:F13,SUM,2,2)
```

Brand -> Sales Channel hierarchy with brand subtotals and a grand total.

Sixth argument: sort_order

In a two-column result, 1 / -1 sort the first column ascending / descending; 2 / -2 sort the second column ascending / descending.

```
=GROUPBY(B2:B13,F2:F13,SUM,2,1,-2)
```

Brand	Net Sales (descending)
GammaFit	15,000
AlphaWear	14,000
BetaStyle	7,000
Grand Total	36,000

Seventh argument: filter_array

```
=GROUPBY(B2:B13,F2:F13,SUM,2,1,-2,C2:C13="Online")
```

Filters source rows to Online transactions before aggregating. Output: AlphaWear 5,000; BetaStyle 4,500; GammaFit 4,500; Grand Total 14,000.

```
=GROUPBY(B2:B13,F2:F13,SUM,2,1,-2,(C2:C13="Online")*(F2:F13>2000))
```

Both conditions must be met. Correct grouped result: AlphaWear 3,000; GammaFit 3,000; BetaStyle 2,500; Grand Total 8,500. Values exactly equal to 2,000 are excluded.

Remember: The filter array must align row-for-row with the grouping and value ranges. Use > for strict greater-than and >= to include the threshold.

Lesson 113 | GROUPBY with HSTACK and field relationships

In the transcript, Month Name and Month Number are nonadjacent source fields. HSTACK brings them together before grouping. The analysis is restricted to VIP customers and ordered chronologically.

Practice columns

Column	Field
B	Month Name
C	Customer Segment
D	Net Sales
F	Month Number

```
=HSTACK(B2:B13,F2:F13)
```

Creates two neighboring columns: Month Name and Month Number, preserving row alignment.

Monthly VIP revenue: complete formula

```
=GROUPBY(HSTACK(B2:B13,F2:F13),D2:D13,SUM,2,1,2,C2:C13="VIP",1)
```

Month	Number	VIP Net Sales
January	1	13,000
February	2	12,500
March	3	13,500
April	4	4,000
May	5	4,500
Grand Total		47,500

Eighth argument: field_relationship

Value	Meaning
0: Hierarchy	Fields form a hierarchy; a later field is sorted within earlier groups
1: Table	Fields are treated as table columns; supports sorting globally by Month Number

The formula uses sort_order=2 (Month Number ascending) and field_relationship=1 (Table mode), giving January-to-May order rather than alphabetical month-name order.

Remember: Table field relationship mode does not support hierarchical subtotals; use a grand total only (total_depth=1).

Final revision | One-page formula reference

Function / topic	Reference formula
TOCOL	=TOCOL(B2:F3,3,TRUE)
TOROW	=TOROW(B2:F3,3,TRUE)
Unique horizontal values	=UNIQUE(TOROW(B2:F3,3,TRUE),TRUE)
VSTACK	=VSTACK(A1:D4,F2:I4)
HSTACK	=HSTACK(A1:C4,E1:F4)
TAKE	=TAKE(A1:J11,5,8)
DROP	=DROP(A1:J11,0,-2)
CHOOSECOLS	=CHOOSECOLS(A1:J11,1,2,7,8,6)
CHOOSEROWS	=CHOOSEROWS(A1:J11,1,3,5,8)
GROUPBY, SUM	=GROUPBY(B2:B13,D2:D13,SUM)
GROUPBY, multiple fields	=GROUPBY(B2:C13,F2:F13,SUM,2,0)
GROUPBY, sort by value	=GROUPBY(B2:B13,F2:F13,SUM,2,1,-2)
GROUPBY, filtered	=GROUPBY(B2:B13,F2:F13,SUM,2,1,-2,C2:C13="Online")

Common troubleshooting

Symptom	What to check
#NAME?	Unsupported function / outdated Excel build
#SPILL!	Clear output cells; ensure output is not inside an Excel Table
#N/A after HSTACK/VSTACK	Input arrays have different heights or widths
Incorrect HSTACK employee mapping	Ensure equivalent row order; HSTACK is not a key-based join
Repeated column headers	Include source header once for VSTACK or choose GROUPBY field_headers deliberately
Unexpected month order	Sort by Month Number with GROUPBY field_relationship=1

End of Lessons 106-113. These notes follow the user-provided lesson sequence, using illustrative practice datasets and checked outputs.